

Swirl, alignment, and the leftover in three-dimensional Navier–Stokes

A map, not a proof

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Abstract

The geometric swirl leftover after Constantin–Fefferman (1993) and Beirão da Veiga–Berselli (2002) is recorded. Alignment of vorticity direction at Lipschitz scale, and then at Hölder exponent 1/2, is a theorem as an if. The pairs those papers refused remain. This note names the missing local bound WRITE (6) and does not prove it. A forced finite-time construction with a smooth external force, announced 8 September 2026, addresses a different official statement (Fefferman (C)/(D)) from unforced regularity ((A)/(B)). It is not WRITE (6). A separate note records global regularity for a Ladyzhenskaya / p -Laplacian augmentation; that is a different equation.

1. What swirl is, in this note

Let u be a divergence-free velocity in three dimensions, $\omega = \text{curl } u$, $\xi = \omega/|\omega|$ off the zero set, and $\varphi(x,y)$ the angle between $\xi(x)$ and $\xi(y)$. Vortex stretching is a principal-value integral of a geometric factor D times $|\omega(y)| / |x-y|^3$, with $|D|$ at most a constant times $|\sin \varphi|$. Local enstrophy on a cutoff supported in a ball of radius r has stretching on the right and $\nu \int |\text{grad } \omega|^2 \varphi$ on the left. This is the classical unaugmented equation. No extra stress. No Φ -cancel of the axisymmetric tube source.

2. The cut that sits, as an if

Constantin–Fefferman (1993): if $|\sin \varphi|$ is at most Lipschitz in $|x-y|$ whenever both vorticities are large, the solution is strong. Beirão da Veiga–Berselli (2002): Hölder exponent 1/2 suffices. Alignment drops the kernel. Call those pairs Good. The complementary intense pairs, with $|\sin \varphi|$ larger than $C_* |x-y|^{1/2}$, are Bad. Good pairs are their theorem. Bad pairs are the pairs they refused. Grujić (2009) localizes the condition, not the bad-pair integral. Later papers weight coherence or assume sparseness. Those are still ifs.

3. WRITE (6)

On a parabolic cylinder Q_r , Bad pairs only, both ends in the local ball:

$$A_{\text{bad}}(Q_r) \leq (\nu/8) \iint |\text{grad } \omega|^2 \varphi + C r^{-2} \iint |\omega|^2.$$

The kernel on Bad is still $|z|^{-3}$. Hardy–Littlewood–Sobolev returns local E^3 , the cubic wall. The Hölder cut changes the set, not the exponent. Assuming the Bad set is thin is the Good-pair theorem again. Morrey: $W^{1,2}$ does not embed in Hölder 1/2 in three dimensions, so energy does not empty the Bad set. $\nu/8$ is a conventional slice of local dissipation, not a sharp constant. Parabolic scaling of the three terms agrees. There is no dimensional obstruction. There is also no proof. Aired leftover: yes. Theorem: no.

4. A cylinder still needs more than (6)

Do not merge the pieces. Good pairs are an if; localizing them leaves a cutoff error on the annulus. Skin flux

at energy level is Caffarelli–Kohn–Nirenberg as smallness, not as an a priori from finite energy. Exterior Biot–Savart is not absorbed as $r \rightarrow 0$ from energy; that remainder is why a localization paper exists. If those bounds and (6) all sit, local Serrin (1962) gives smoothness on a smaller cylinder. That implication is not Caffarelli–Kohn–Nirenberg. If (6) sits and skin flux from energy does not, the cylinder is still open. If the cylinder is already small in the Caffarelli–Kohn–Nirenberg sense, (6) was not needed on that cylinder. Nečas–Růžička–Šverák (1996) rules out Leray backward self-similar profiles. It is not a blanket Type-I theorem.

5. A forced vortex is not this leftover

Fefferman’s official statement has four alternatives. (A) and (B): unforced, smooth data stay smooth on $\mathbb{R}^3$ or on the torus. (C) and (D): a smooth force is allowed, and one asks for breakdown. On 8 September 2026 a finite-time singularity with a smooth external force was announced, pictured as a vortex that spirals inward. If that write-up holds, its slot is (C) or (D). It does not decide (A) or (B). It does not bound WRITE (6). I did not write that construction. I do not claim it. This note is a map of the unforced geometric leftover. The problem those symbols name is 1993. The naming of WRITE (6) in this form is 10 September 2026.

6. What this author claims, and what he does not

Claims: the map above; a separate write-up of global regularity for Navier–Stokes plus Ladyzhenskaya / p-Laplacian stress Q_1 at $\varepsilon > 0$, $\beta \geq 1/2$, on the three-torus (this PDE only; known class; cite Ladyzhenskaya 1968/1969 and Málek–Nečas–Růžička 1996); inverse-GCD matrix facts posted as Zenodo 22045478, not as the Riemann hypothesis. Does not claim: ordinary unforced Navier–Stokes; axisymmetric ordinary NS with swirl as finished; uniform H^1 as $\varepsilon \rightarrow 0$ for the augmented equation; WRITE (6) as a theorem; the forced construction of 8 September 2026. Axisymmetric swirl, keeping $1/r^4$: a localized Hardy inequality does not absorb all data. A Φ -cancel of the tube source was dropped; it was not the unforced path.

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